Your First Database: Docker, Postgres and Python



GOLDEN GATE
UNIVERSITY

Lab: DBMS and network setup, Python setup, summary statistics

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What we will do today

15
min
Start a database
with Docker

15
min
Connect
with Python

25
min
Summary statistics
in SQL and Python

5
min
Wrap up

Goal: No database experience needed.

The big picture



Your laptop

Goal: A host name (localhost) plus a port number (5432) is the address of the database.

Useful Links

Docker With PostgrSQL:

https://www.youtube.com/watch?v=RdPYA-wDhTA

Download Python on Windows:

https://www.youtube.com/watch?v=ddGTXBhaGWA

Before you start

1. **Docker Desktop**

```
docker --version
```

2. **Python 3.10 or newer**

```
python --version
```

3. **Lab folder with 2 files**

```
instructor.csv, load_data.py
```

Run the commands in a terminal (Windows: PowerShell). Each should print a version number.

Goal: Three things ready: Docker, Python, and the lab folder.

Step 1: Start the database

```
docker run --name labdb -e POSTGRES_PASSWORD=lab -p 5433:5432 -d postgres:16
```

`--name labdb`	a name for our container
`-e POSTGRES_PASSWORD=lab`	the password is "lab"
`-p 5432:5432`	open port 5432 to our laptop
`-d postgres:16`	run Postgres version 16 in the background

Goal: One command gives you a running Postgres server. No installer needed.

Step 2: Check it works and create a database

```
docker ps

docker exec -it labdb psql -U postgres

CREATE DATABASE lab;
\c lab
SELECT version();
\q
```

What you should see

- docker ps lists labdb as Up
- The postgres=# prompt appears
- CREATE DATABASE is printed
- version() shows PostgreSQL 16
- \q leaves psql

Goal: A running server with an empty database called lab.

Step 3: Set up Python

```
Use Windows Powershell
1) python -m venv labenv
2) Set-ExecutionPolicy -Scope Process RemoteSigned
3) pip install pandas sqlalchemy psycopg2-binary matplotlib
```

pandas
tables and statistics

sqlalchemy + psycopg2
talk to Postgres

matplotlib
draw charts

Goal: Python with the tools we need. The prompt shows (labenv) when it is active.

Step 4: Load the data from Python

```
python load_data.py
```

It prints: Loaded 100 rows

The data

id	name	department	salary
1	Asha Das	Biology	73500
2	Ben Khan	Physics	57500
3	Chen Singh	History	79000

Check the load in psql

```
docker exec -it labdb psql -U postgres -d lab

SELECT COUNT(*) FROM instructor;
-- 100

SELECT * FROM instructor LIMIT 3;
```

Goal: The table instructor exists in Postgres and has 100 rows.

Step 5: Summary statistics in SQL

```
SELECT COUNT(*) AS n,
       ROUND(AVG(salary), 2) AS mean,
       ROUND(STDDEV_SAMP(salary), 2) AS sd,
       MIN(salary) AS min,
       MAX(salary) AS max
FROM instructor;

SELECT percentile_cont(ARRAY[0.25, 0.5, 0.75])
       WITHIN GROUP (ORDER BY salary)
FROM instructor;
```

You should get

n	**100**
mean	**71,960**
sd	**19,003.55**
min / max	**46,500 / 180,000**
Q1 / median / Q3	**63,125 / 70,250 / 77,125**

Goal: The database does the maths and sends back just a few numbers.

Step 6: The same statistics in Python

```
1) notepad stats.py

//stats.py
import pandas as pd
from sqlalchemy import create_engine
import matplotlib.pyplot as plt

url = "postgresql+psycopg2://postgres:lab@localhost:5432/lab"
engine = create_engine(url)
df = pd.read_sql("SELECT * FROM instructor", engine)

s = df["salary"]
print(s.describe())
print(s.median())
```

Do SQL and Python agree?

Compare mean, median and standard deviation with Step 5.

They should match.

Goal: Same data, two tools, same answers. That is how we check our work.

Step 7: Draw it and spot outliers

```
q1, q3 = s.quantile([0.25, 0.75])
iqr = q3 - q1
print(s[(s < q1 - 1.5*iqr) | (s > q3 + 1.5*iqr)])

fig, ax = plt.subplots(1, 2, figsize=(9, 3.5))
ax[0].hist(s, bins=10)
ax[1].boxplot(s, vert=False)
plt.savefig("salary_plots.png")

2) python stats.py
3) start salary_plots.png
```

Expected

3 outliers:
150,000, 165,000 and 180,000

- Histogram leans right
- Boxplot shows dots far to the right
- An outlier is a question to ask, not a value to delete

Goal: Read the summary numbers and the picture together.

If something goes wrong

Symptom	What to do
docker: command not found, or cannot connect to daemon	Open Docker Desktop and wait until it says Running
port is already allocated	Use -p 5433:5432 and port 5433 in the Python URL
connection refused in Python	Run docker ps. If labdb is missing: docker start labdb
password authentication failed	The password is lab, user is postgres
relation "instructor" does not exist	Run python load_data.py again

Goal: Read the error message first. It usually tells you what is wrong.

Wrap-up and hand-in

Hand in

- Screenshot of docker ps and the SELECT COUNT(*) result
- Your SQL and Python statistics side by side
- salary_plots.png

Clean up

```
docker stop labdb
docker rm labdb
```

This stops and removes the database. Your Python files stay.

Goal: You ran a database server and queried it from Python.